

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error

df = pd.read_excel(r"C:\Users\harsh\OneDrive\Desktop\AICONCEPT\
supply_chain_sales.xlsx")
print(df.head())
```

	Date	Product	Sales	Price	Promotion	Holiday
Temperature						
0	2024-01-01	Product A	119	490	0	0
20						
1	2024-01-02	Product A	125	500	0	0
20						
2	2024-01-03	Product A	139	500	1	0
20						
3	2024-01-04	Product A	126	500	0	0
21						
4	2024-01-05	Product A	113	490	0	0
25						

```
df["Date"] = pd.to_datetime(df["Date"])
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 365 entries, 0 to 364
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Date            365 non-null   datetime64[us]
1   Product         365 non-null   str
2   Sales           365 non-null   int64
3   Price           365 non-null   int64
4   Promotion       365 non-null   int64
5   Holiday         365 non-null   int64
6   Temperature     365 non-null   int64
dtypes: datetime64[us](1), int64(5), str(1)
memory usage: 20.1 KB
```

```
print(df.dtypes)
```

```
Date            datetime64[us]
Product         str
Sales           int64
Price           int64
Promotion       int64
Holiday         int64
Temperature     int64
dtype: object
```

```
# Feature engineering
```

```
df['Day'] = df['Date'].dt.day
df['Month'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year
df
```

```
      Date  Product  Sales  Price  Promotion  Holiday
Temperature Day \
0  2024-01-01  Product A    119    490         0         0
20    1
1  2024-01-02  Product A    125    500         0         0
20    2
2  2024-01-03  Product A    139    500         1         0
20    3
3  2024-01-04  Product A    126    500         0         0
21    4
4  2024-01-05  Product A    113    490         0         0
25    5
..      ...      ...      ...      ...      ...      ...
..      ...
360 2024-12-26  Product A    105    510         0         0
20    26
361 2024-12-27  Product A    115    510         0         0
20    27
362 2024-12-28  Product A    116    500         0         0
22    28
363 2024-12-29  Product A    117    510         0         1
21    29
364 2024-12-30  Product A    108    490         0         0
20    30

      Month  Year
0         1  2024
1         1  2024
2         1  2024
3         1  2024
4         1  2024
..      ...  ...
360      12  2024
361      12  2024
362      12  2024
363      12  2024
364      12  2024

[365 rows x 10 columns]

df.isnull().sum()
```

```

# remove duplicate rows if any
df = df.drop_duplicates()

# duplicates already removed by df = df.drop_duplicates()

# features and target
X = df[['Day', 'Month', 'Year']]
y = df['Sales']

```

train test split

```

from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(x_train, y_train)

LinearRegression()

```

prededctions

```

predictions = model.predict(x_test)
mae = mean_absolute_error(y_test, predictions)
print("Mean Absolute Error:", round(mae, 2))

```

Mean Absolute Error: 10.52

predict future demand

```

future = pd.DataFrame({
    'day': [11, 12, 13],
    'month': [1, 1, 1],
    'year': [2024, 2024, 2024]
})

future_predictions = model.predict(future)

print("Future demand predictions:")
print(future_predictions)

```

Future demand predictions:
[130.21491095 130.1007789 129.98664685]

```

future_pred = model.predict(future)

```

```

print("\nFuture Demand Prediction")

for day, demand in zip(future['Day'], future_pred):
    print(f"Day {day}: {round(demand)} units")

Future Demand Prediction
Day 11: 130 units
Day 12: 130 units
Day 13: 130 units

#demand optimizes
current_inventory=120
predicted_demand=185
safety_stock=30
reorder_quantity=predicted_demand+safety_stock-current_inventory
if reorder_quantity>0:
    print("order quantity",reorder_quantity,"units")
else:
    print("inventory is sufficient")

order quantity 95 units

```